Evaluating AI-based information search tools using the TRUST framework

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It's hard to review search tools, especially if they incorporate AI.

Search tools are rapidly being integrated with new search technologies like AI. Existing approaches to evaluate these tools are often complex, unstructured, and/or time consuming. There is a need for a structured and user-friendly method to assess reliability, transparency, ethics, and scholarly fit of search tools. Librarians can play a key role in the creation and use of evaluation methods that streamline this process to give the best advice and support on the use of these tools.

The **TRUST** framework combines a structured set of questions & minimum requirements to assess (AI-based or other) search tools. We implemented this as a browser plugin that lets users evaluate tools directly from their own browser fully locally and free. The plugin lets users capture evidence (webpage snapshots), score with argumentations, log structured metadata, and export the results as a standalone nicely formatted report.

We built a framework + browser extension to review search tools.

Demo, install instructions, and more

https://trust.samuelmok.cc

PRIVATE | IP SAFE | ACCESSIBLE | SOURCED | SECURE

TRUST

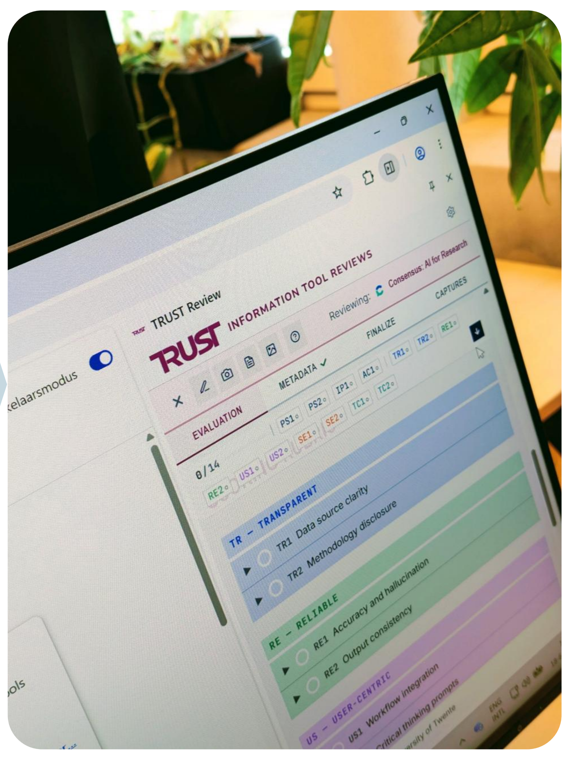
TRANSPARANT: TR1 SOURCE CLARITY, TR2 METHODOLOGY
RELIABLE: RE1 ACCURACY, RE2 CONSISTENCY, RE3 HALLUCINATION
USEABLE: US1 INTEROPERABLE, US2 CRITICAL THINKING
SOUND: SO1 FAIRNESS, SO2 DATA USE
TRACEABLE: TC1 ATTRIBUTION, TC2 QUALITY INDICATORS

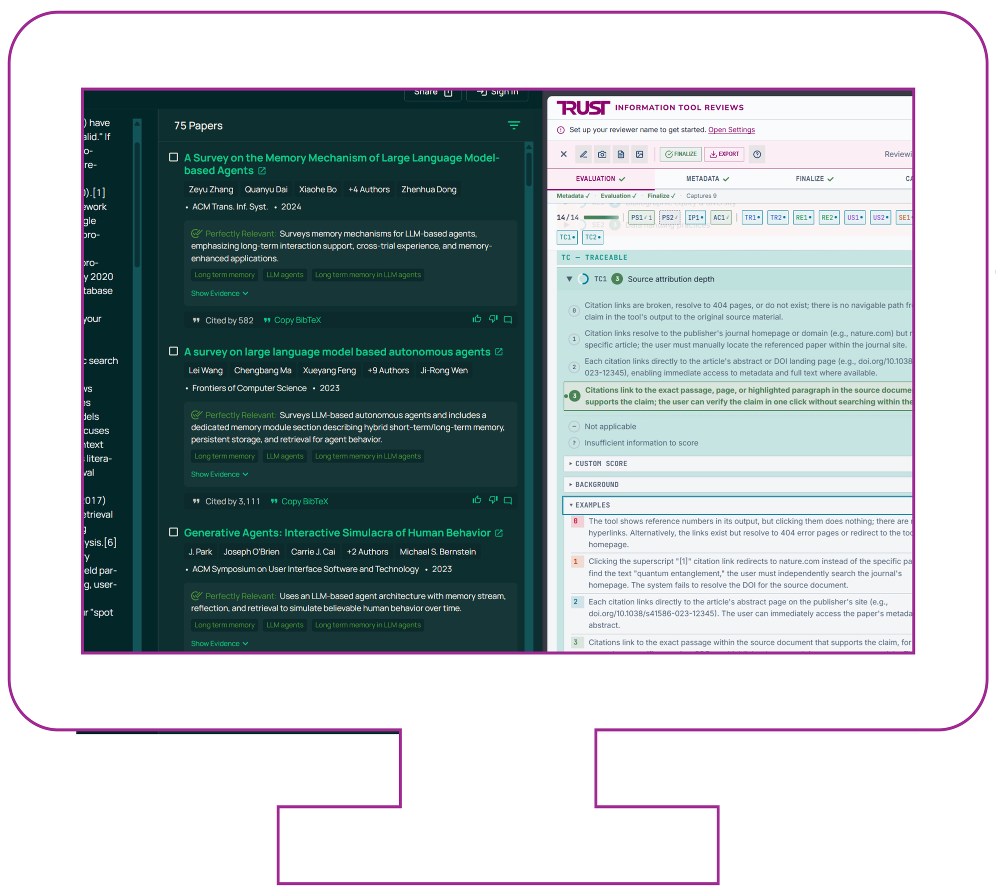
The framework is a set of questions clustered into 5 subjects TR RE US SO TR, plus separate pass/fail requirements on top.

Each question has a fully defined rubric, example scores, and background information.

The browser extension is custom built to allow users to work with the framework directly in their own browser.

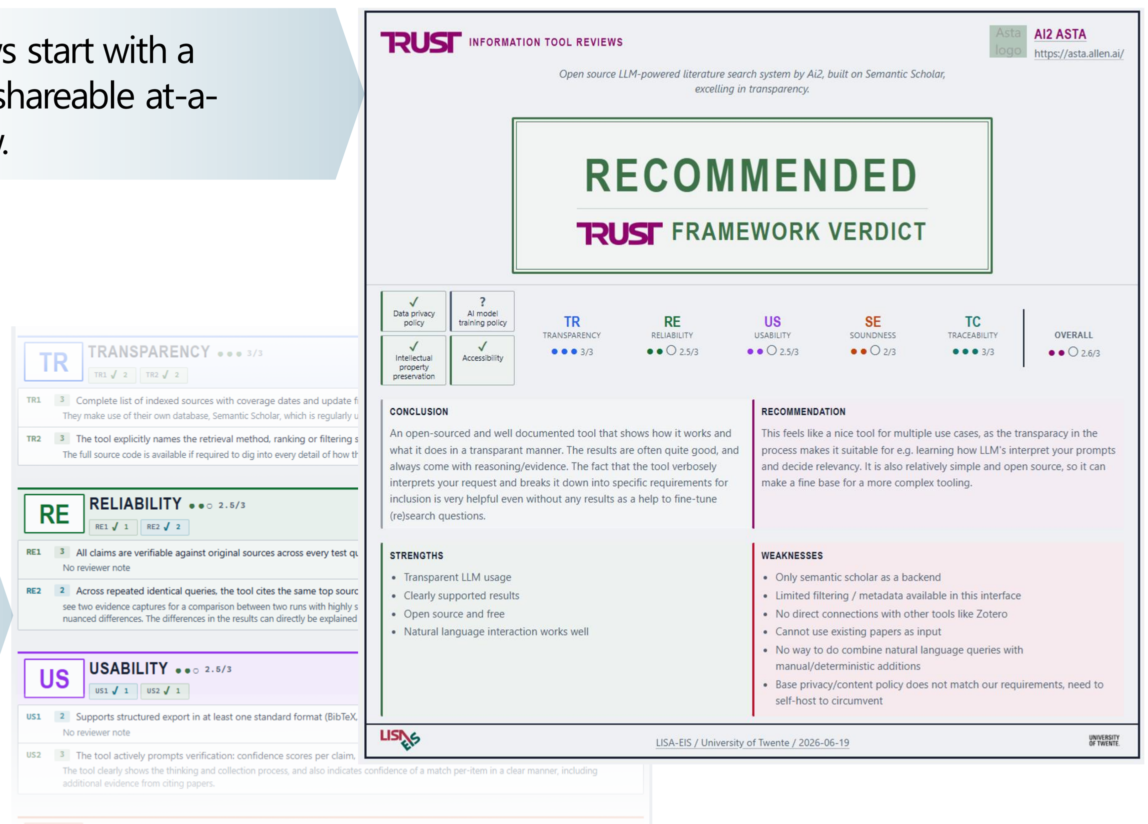
The extension has all questions & details directly built in, plus tools like creating web snapshots and exporting reports. It's free, open source, and easily installed from the Chrome extension store.





The extension is a browser sidebar. This allows users to review tools in a known environment without the need for switching screens.

Exported reviews start with a shareable at-a-glance overview.



Details are also covered in the report, including scores, screengrabs, notes, metadata, ...

Reviews are a step in a process.

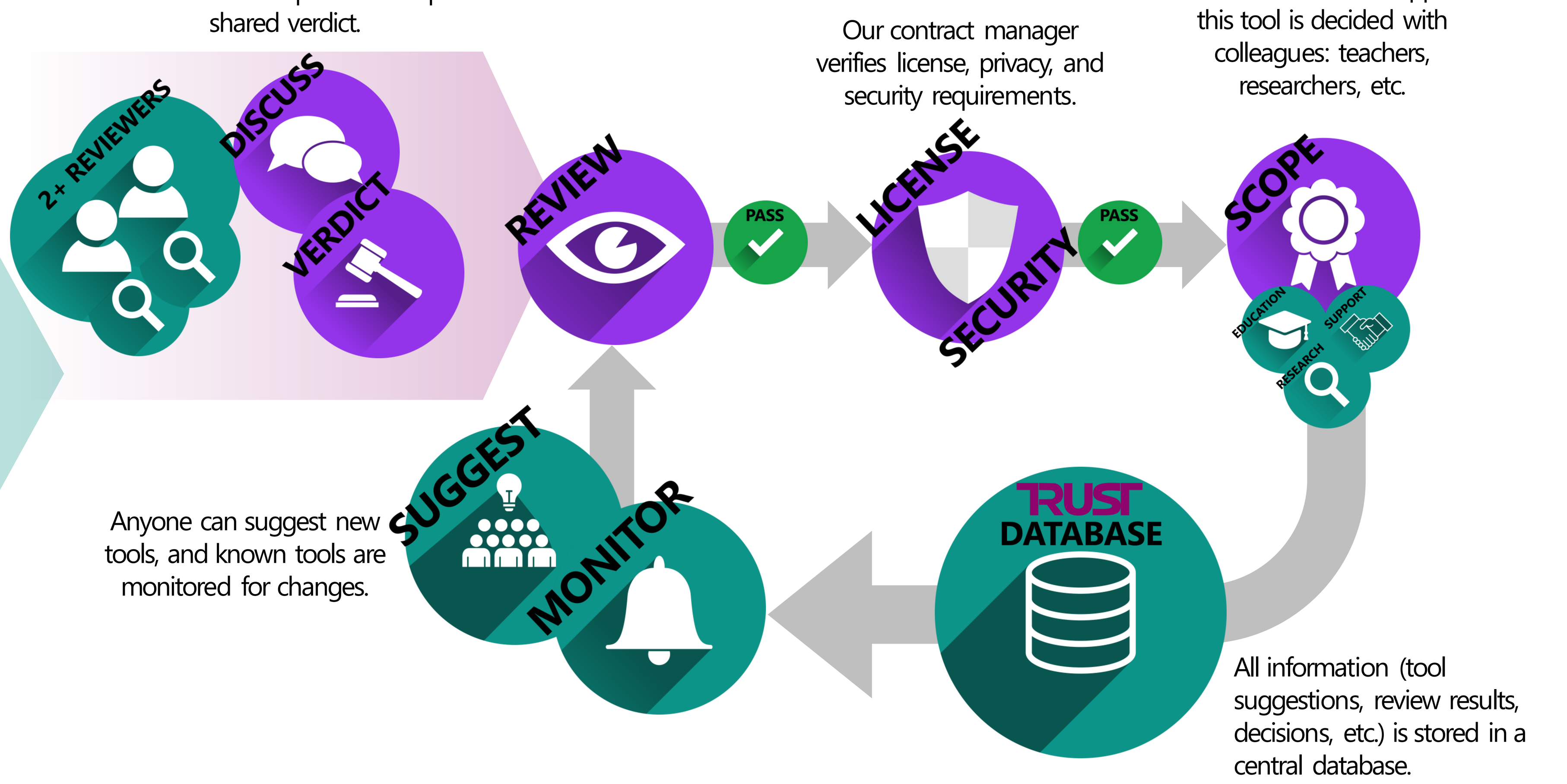
The review process is meant to be embedded into a larger workflow that matches organizational requirements. We are currently working on implementing a process like shown to the right. Some steps are outside our expertise, such as licensing checks. These will require collaboration with relevant departments, such as contract management and the AI office.

The review extension has been designed to be easily modified to fit specific needs for workflows or review frameworks: the question rubric can be changed with a simple text editor, and the full source code is available under an open license if needed.

Reviews are done in pairs and require a shared verdict.

Our contract manager verifies license, privacy, and security requirements.

Where and how to support this tool is decided with colleagues: teachers, researchers, etc.



Anyone can suggest new tools, and known tools are monitored for changes.

All information (tool suggestions, review results, decisions, etc.) is stored in a central database.

Example of tooling in development. Reviews can be compared side by side, e.g. for reaching consensus.

COMPARE TOOLS

CRITERION	Ai2 Asta	Undermind
Verdict	An open-sourced and well documented tool that shows how it works and what it does in a transparant manner. The results are often quite good, and always come with reasoning/evidence. The fact that the tool verbosely interprets your request and breaks it down into specific requirements for inclusion is very helpful even without any results as a help to fine-tune (re)search questions.	Undermind looks clean and is user-friendly. The clarifying questions before starting the search are very helpful. However, their documentation is weak and it is a real black box, with no reproducibility nor good quality indicators of the sources it reports. Do not use as the only AI tool.
Score	23/27	11/27
● Transparency	3.0	1.5
● Reliability	2.5	1.5
● Usability	2.5	0.5
● Soundness	2.0	1.0
● Traceability	3.0	1.5
Strengths	• Transparent LLM usage • Clearly supported results • Open source and free • Natural language interaction works well	• Asks clarifying questions before the search starts • Continuous chat function • Chat function about specific papers • Relevance ranking including why it is relevant to the research question • Extensive report based on substantial amount of sources
Weaknesses	• Only semantic scholar as a backend • Limited filtering / metadata available in this interface • No direct connections with other tools like Zotero • Cannot use existing papers as input • No way to do combine natural language queries with manual/deterministic additions • Base privacy/content policy does not match our requirements, need to self-host to circumvent	• No or hardly finable documenation • Generic privacy policy and terms of use • Weak reproducibility • Free version based on abstracts, premium includes open access • Login required

Where do we go from here?

- ❑ Develop a central database to store and manage all evaluations.
- ❑ Establish monitoring process to keep reviews up to date and trigger re-evaluation when needed.
- ❑ Create "golden datasets" of queries + expected results, to be used while reviewing tools to do objective comparisons.
- ❑ Refine and implement the workflow in collaboration with relevant departments.
- ❑ Fully integrate workflow into our day-to-day work, e.g., in outreach, education, and research support.
- ❑ Putting our own tool into practice beyond this pilot.